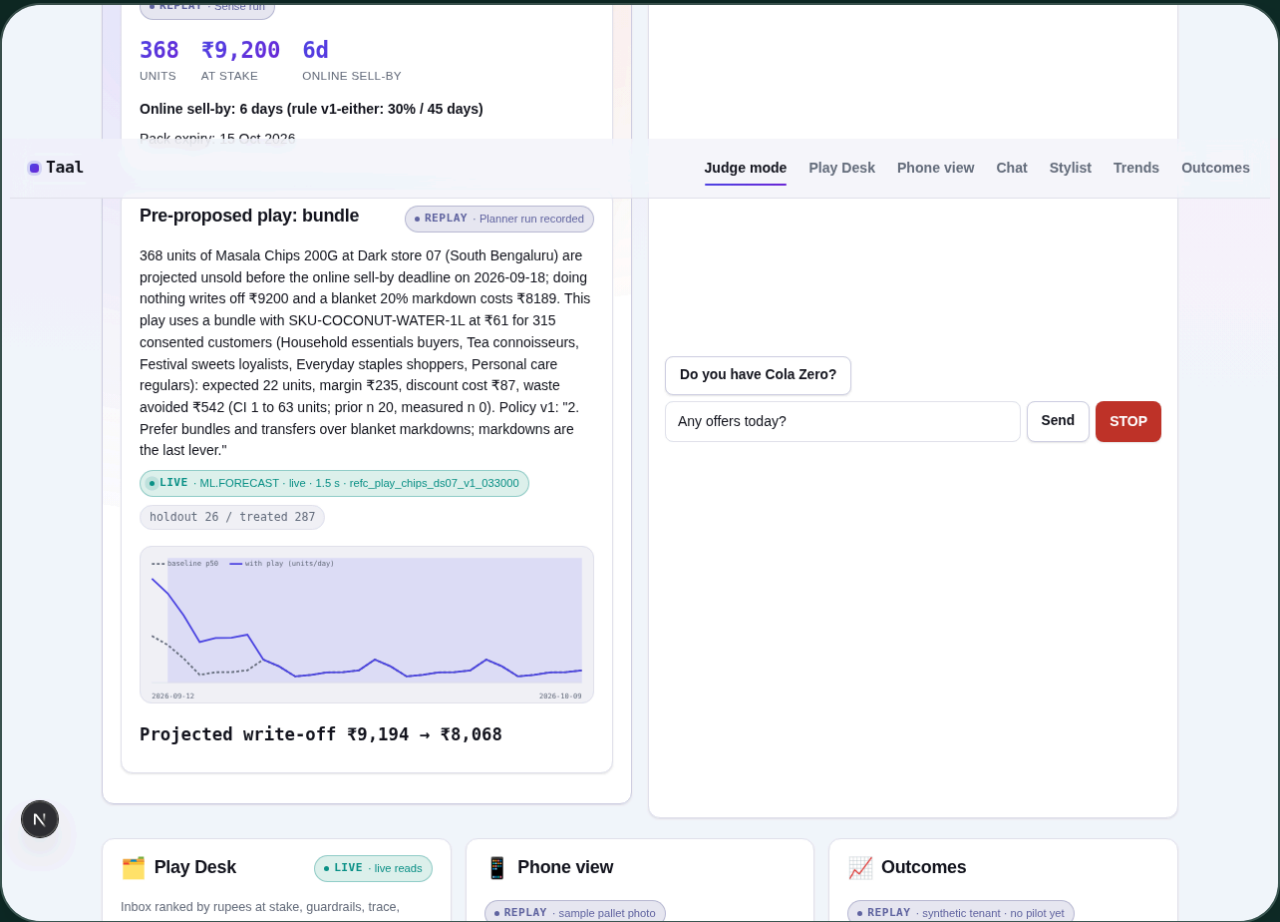


Taal

The agent that sells what the forecast says
you'll throw away.

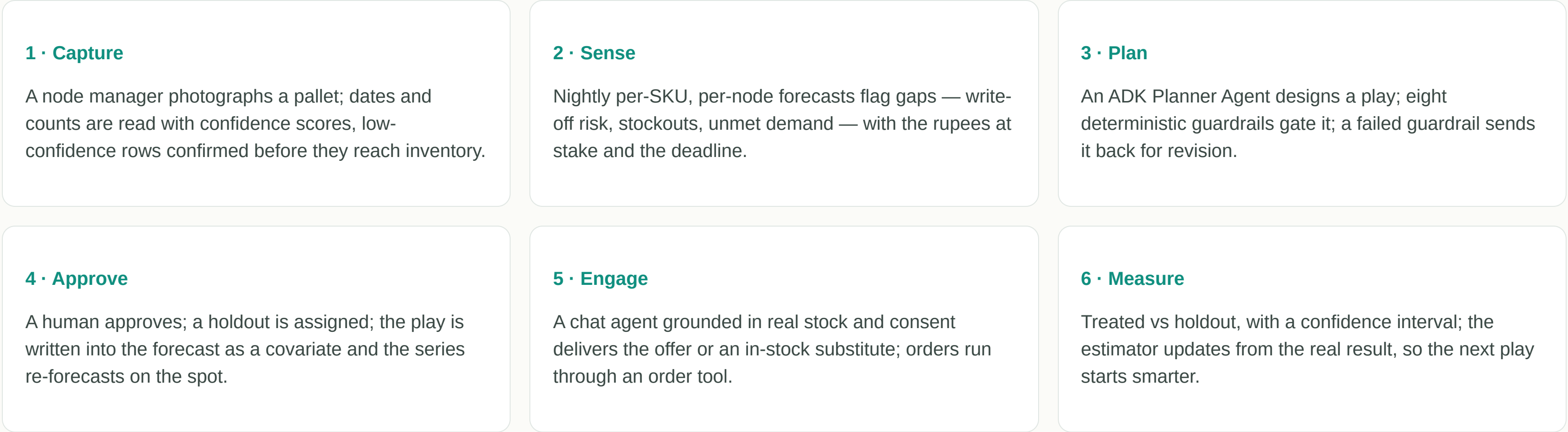
Kutumb Mart · 10 dark stores, 6 outlets, Bengaluru

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WHAT IT DOES

Six steps from a forecast to a sale



ONE REAL RUN, THIS TENANT ₹9,194 → ₹8,068 projected write-off on one lot, before and after a live guardrail-approved play

THE HOOK

33 days on the pack. 6 days to sell it online.

India's food regulator requires 30% of shelf life or 45 days remaining at delivery for food sold online. No forecasting tool knows that deadline exists — Taal computes it per lot, versioned as a tenant policy, and shows it on every gap card.

Demo lot, dark store DS-07

368 units

Masala Chips · ₹9,200 at stake

INNOVATION

One Play schema, two different problems

A **Play** is one object — target, mechanic, audience, guardrails, holdout — and the gate that checks it (8 guardrails plus the estimator) has zero code specific to either domain below. The same machinery runs both.

GROCERY

A forecast says a lot will go unsold before a deadline

5 gap types: online sell-by, expiry, stockout, rebalance, slow mover

APPAREL

A real customer asks for a style the node doesn't stock

2 gap types: unmet demand, assortment gap — sourced from real chat asks, not a forecast

Both land on the same Planner, the same 8 guardrails, the same holdout-measured loop — see Architecture for how a Play moves through it.

TECHNICAL MERIT

Real services, measured live

Sense (nightly, local model)

BigQuery ARIMA_PLUS_XREG SQL written, fixed, verified on one real series → gaps, segments, substitutes



Planner (ADK LoopAgent, Gemini Flash)

6 tools, max 3 iterations, revises on guardrail failure



Approve (Cloud Run)

holdout assignment, writes future_regressors, re-forecasts



Customer / Stylist Agent + Measure

Gemini chat, MCP order tool, treated-vs-holdout lift

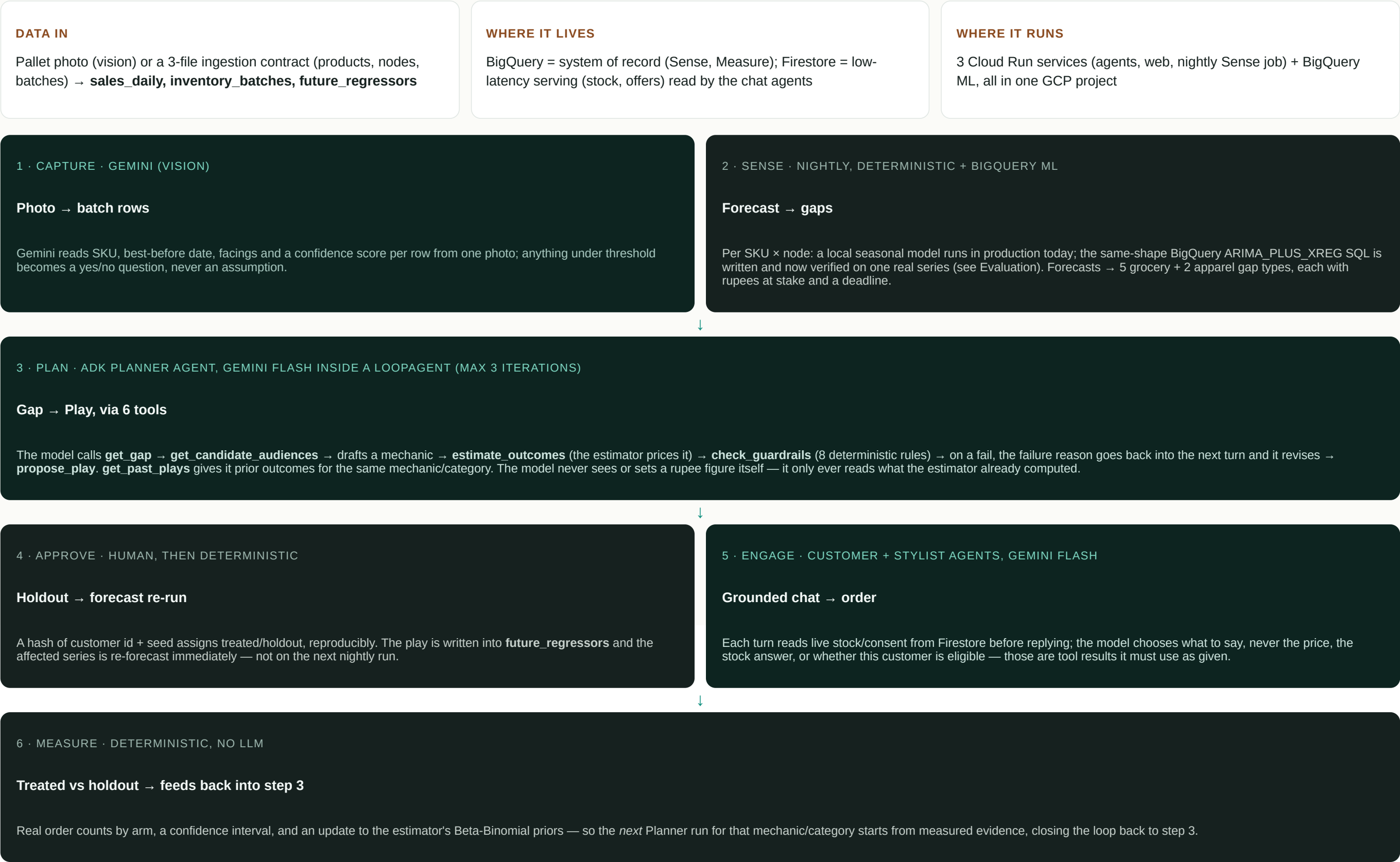
Measured, live Vertex — not estimated

Path	Latency
Customer chat, p50	4.40 s
Customer chat, p95	5.75 s
Approve → re-forecast	8.30 s
Sense, 300 SKUs × 16 nodes	3.83 s

50 real chat calls against Gemini 2.5 Flash on Vertex, no caching, a fresh session per call. Full raw output committed to the repo, not typed by hand.

ARCHITECTURE

How data, agents and models fit together



What Gemini decides — and what it is never allowed to

Gemini decides

- Which mechanic and audience for a gap, and the rationale
- Whether to revise a draft after a failed guardrail
- Reading a pallet photo into SKU, date, count, confidence
- Reading a garment or selfie photo into style attributes
- Dialogue: substitution reasoning, the reply envelope
- Vernacular copy for approved plays (falls back to templates)

Deliberately never LLM

- Every rupee — the estimator, a pure function
- All 8 guardrails — margin floor, consent, holdout, ...
- Holdout assignment — a hash, reproducible from seed
- Stock, offer eligibility, order prices
- Lift, confidence intervals, measurement

The trace panel shows this live: a guardrail returning **passed: false**, and the exact next tool call the Planner makes to fix it — not a black box.

What makes this a decision engine, not a promo bot

A legal deadline as a model input

The FSSAI online sell-by rule is computed per lot and, once a play is approved, written straight into the forecast as a covariate — the series re-forecasts on the spot, not on the next nightly run.

One schema, two domains

Grocery write-off risk and apparel unmet-demand run through the identical Play object, the same eight guardrails and the same estimator — zero domain-specific code in the gate layer.

An estimator that learns

Every Measure run updates the estimator's priors from real treated-vs-holdout counts, so the next play of that shape starts from evidence, not a static assumption.

Not a fourth marketing channel

A decision layer: it emits an audience, copy and price a retailer's existing CRM or channel can send. It decides supply-side, per lot × node × deadline — not campaign scheduling.

Grounded, not general chat

The Customer Agent is grounded in this node's real stock and consent; it never promises a product the stock tool did not confirm.

PROBLEM ALIGNMENT & IMPACT

Three ways to handle the same lot

368 units of Masala Chips, dark store DS-07, 6 days to online sell-by — the same lot from the opening slide

DO NOTHING ₹9,200 written off; estimator projection at plan time	BLANKET 20% MARKDOWN ₹8,189 discounted for everyone, including customers who'd have paid full price; estimator counterfactual	THIS PLAY, LIVE-APPROVED ₹9,194 → ₹8,068 targeted to 287 consented customers, real re-forecast, not simulated
Why now The FSSAI online sell-by rule already exists; quick-commerce and dark-store growth make the online window the binding constraint, not the shelf.	Consent-native, not a growth hack Every audience is filtered by marketing consent before a play exists; STOP withdraws it immediately, checked on every future play, not just at signup.	

EVALUATION

Numbers we can defend

Metric	Result	Note
Agent Simulation, guardrail pass	190 / 190	live Vertex, 5 guardrails, 190 personas
Vision read accuracy, 30 photos	30 / 30	synthetic photos — not a real-camera test
Customer chat p50 / p95	4.40 s / 5.75 s	real gemini-2.5-flash, 50/50 live calls OK
Forecast backtest MAPE, 9 tiers	0.107–0.221	7 rolling origins, 63 rows
Sense throughput	3.83 s	300 SKUs × 16 nodes, one run
BigQuery ARIMA_PLUS_XREG, one series	28/28 rows	real job, 2 real SQL bugs found & fixed running it
Pilot: treated vs holdout lift	not yet measured	no pilot has run
Cost per play, billing export	not yet measured	list-price estimate only, labelled

9 of 13 §12 rows carry a real, committed measurement; the rest say so rather than guess. Every number above is parsed from raw output under **eval/raw/**, never typed by hand.

LONGEVITY

Scalability and sustainability

Throughput, measured

300 SKUs × 16 nodes senses in 3.83 s. Cloud Tasks fan-out and per-gap triage by rupees-at-stake are the extrapolation path to a 20,000-SKU retailer.

Unit economics

Flash-only, no idle endpoints, nightly batch. A Cost Governor gates which gaps get a model call at all, by rupees at stake.

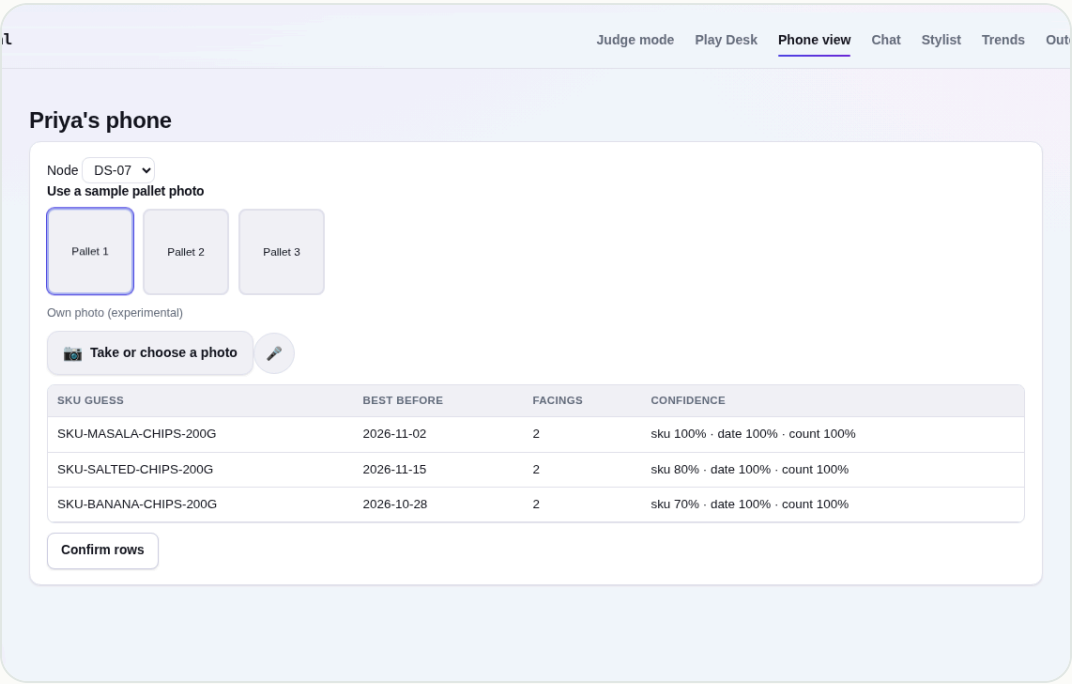
Onboarding

A three-CSV ingestion contract (products, nodes, batches) is the path from a seeded demo tenant to a real retailer's own catalogue.

Multi-tenant from day one

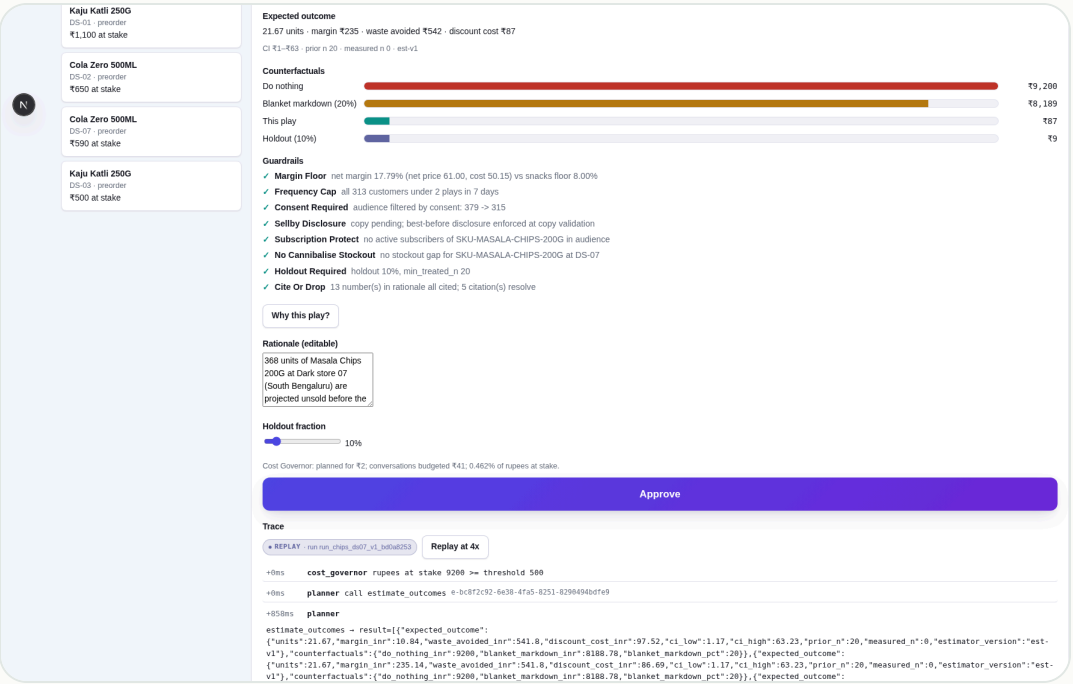
tenant_id on every table, per-tenant policy text — not a single-demo hack. Least-privilege IAM per service is written, not yet wired into the live deploy.

Three people, three screens, one job each



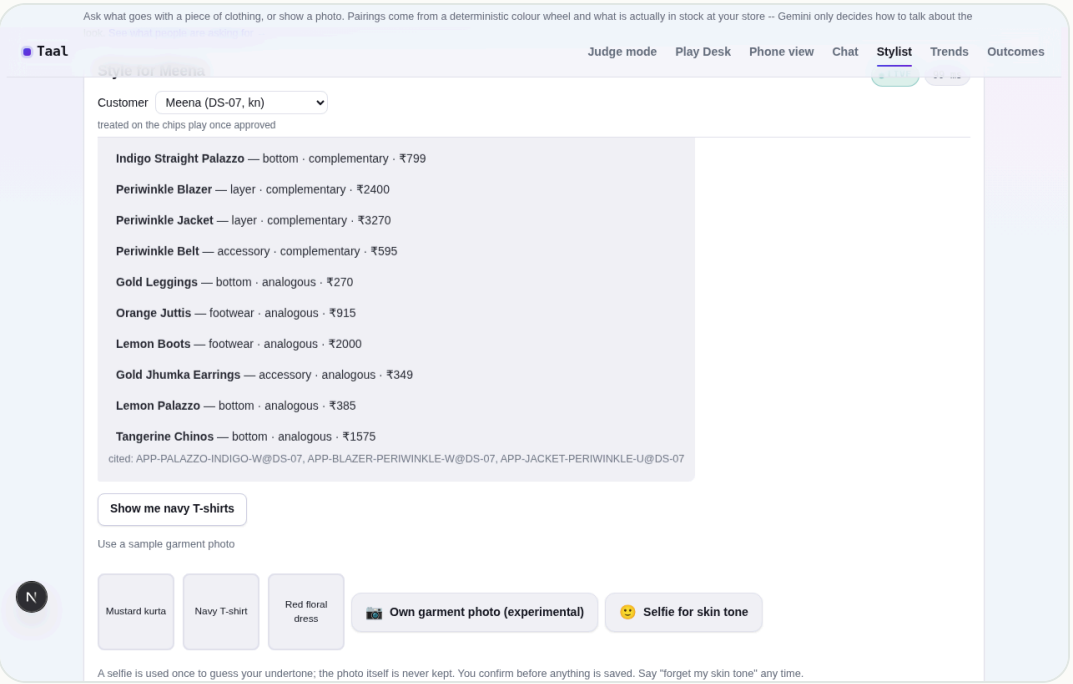
Priya, node manager

Photographs a pallet; low-confidence rows are confirmed, never assumed, before they reach inventory.



Arjun, demand planner

Reviews the rationale, the guardrails, the trace of a failed-then-revised draft, and approves.



Meena, customer

Gets the offer in her language with the best-before date, or an in-stock substitute, never a promise the stock tool didn't confirm.

CLOSING

Every forecast tells a retailer what will be thrown away.

Taal is the agent that sells it first.

REAL

Forecasts (local model; BigQuery SQL written and verified on one series), gaps, estimator, guardrails, the Planner loop, holdout assignment, live re-forecast, chat, orders, measurement

SIMULATED, DISCLOSED

The tenant Kutumb Mart (seeded generator, its rule stated in code); no pilot has run yet — the Outcomes screen says so on every row

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github.com/Nandish3010/probable-fortnight